



GeoDF-Adaptive module (front-end only; no IMU / no CNN)

- (a) Undistort & lift tracks; map to VINS pseudo-pixels
- (b) Temporal 2D-F on left cam (RANSAC) + Sampson residual e_i (support / fallback; also gates 3D activation)
- (c) Stereo: KLT cur left $\rightarrow$ right; triangulate prev stereo (L/R at $t-1$)
- (d) Primary gate: PnP-RANSAC rigid motion $\rightarrow$ 3D reprojection r_i
Dynamic candidate if $r_i > \tau_{3D}$ (else 2D dual-gate fallback)
- (e) Scene-aware ARM: $\text{EMA}(\text{outlier ratio}) \geq \rho_{\text{on}}(\text{auto})$ AND quality gate
($\rho_{\text{on}} = \text{clamp}(\text{floor} \cdot 1.8 + 0.10, 0.14, 0.40)$; hysteresis disarm)
- (f) Temporal voting: flag $\geq k=3$ consecutive frames (warmup 60)
Ratio guard: reject $\leq 40\%$ tracks, cap 5 deletions / frame
- (g) If ARMED: hard-delete voted tracks; else pass-through all features

Insertion point: after temporal KLT, before setMask() — back-end estimator unchanged.